

The Water Cycle

Chapter 1: Overview and Key Processes

The water cycle, also known as the hydrological cycle, describes the continuous movement of water on, above, and below the surface of Earth.

Evaporation is the process by which liquid water is converted into water vapor. It occurs when energy from the sun heats the surface of oceans, lakes, and rivers. Approximately 86% of global evaporation occurs from the ocean surface.

Transpiration is the process by which plants release water vapor through their leaves. The combined process of evaporation and transpiration is called evapotranspiration.

Condensation occurs when water vapor cools and transforms into liquid droplets, forming clouds and fog. This process requires condensation nuclei, which are tiny particles like dust or sea salt around which water droplets form.

Precipitation is water released from clouds as rain, sleet, snow, or hail. It is the primary mechanism by which water is delivered from the atmosphere back to the Earth's surface.

Runoff is the flow of water over the land surface into streams and rivers. It occurs when precipitation exceeds the soil's infiltration capacity.

Infiltration is the process by which water seeps into the soil, eventually reaching groundwater aquifers that can supply wells and springs.

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Chapter 2: Storage and Global Significance

Water Storage Reservoirs

Oceans store approximately 96.5% of all Earth's water. The residence time of water in the ocean averages about 3,100 years.

Glaciers and ice caps hold about 1.74% of Earth's total water. They represent the largest reservoir of fresh water on Earth, storing water for thousands of years.

Groundwater accounts for roughly 1.69% of all water and can be accessed through wells and springs. It plays a critical role in sustaining ecosystems and human water supplies during dry seasons.

Lakes and rivers contain only about 0.013% of total water but are the primary source of freshwater for human consumption and agriculture.

The Solar Energy Driver

The water cycle is primarily driven by solar energy. The sun provides the heat needed for evaporation. Wind patterns, shaped by differential solar heating, transport moisture across continents. Gravity drives precipitation and runoff back toward lower elevations and ultimately the ocean.

Climate Feedback Loops

Water vapor is the most abundant greenhouse gas in the atmosphere, amplifying the warming effects of carbon dioxide. As temperatures rise, more evaporation occurs, increasing atmospheric water vapor and further warming the planet. This constitutes a positive feedback loop within the climate system.

Human Impacts

Deforestation reduces transpiration, altering local precipitation patterns. Urbanization increases impervious surfaces, increasing runoff and reducing groundwater recharge. Irrigation diverts water from natural flow paths. Over-extraction of groundwater leads to aquifer depletion worldwide.